

Lucian Medrihan

Software Engineer

About Me

Passionate Software Engineer with Full Stack development experience, which began by exploring Ethical Hacking at age 12. My career spans from GIS applications to management systems, with a focus on machine learning projects. I have contributed to projects for SIAN and ESA, demonstrating strong problem-solving capabilities.

Contact

+39 3272898921
medrihanlucian@gmail.com
lvcian.github.io/Portfolio
linkedin.com/in/lume
github.com/lvcian
Rome, Italy

Skills

Technical Skills: Proficient in **Python**, JavaScript, and Node.js, with extensive experience in **React.js**, **Angular.js**, Flask, and FastAPI. Expertise in **full-stack development** using modern web technologies, complemented by strong skills in **Docker**, **AWS**, and cloud solutions.

Additional Skills: **GIS**, **Machine Learning**, **LLMs**, AI-assisted development, Image Processing, Tailwind CSS, Bootstrap, Openlayers, Linux commands, **Cybersecurity**, Geopandas, JQuery, network programming.

Soft Skills: Proven ability to solve **complex problems**, manage projects effectively, and **lead teams**. Strong time management, critical thinking, and attention to detail.

Education

● **Computer and Automation Engineering**
Sapienza University of Rome

Languages

Italian	★ ★ ★ ★ ★
Romanian	★ ★ ★ ★ ★
English	★ ★ ★ ★ ★

Achievements & Activities

- **Bug Bounty Hunter:** Found 2 critical RCEs (CVSS 10.0) on major Italian ISP via Cyberdart.eu (\$5k bounties)
- **Google Workshop Winner** for Uni-verCity, presented at Google Tel Aviv
- Co-Founder of **Hackappatoi**, Sapienza's top ethical hacking / CTF team

Experiences

● **Full Stack Software Engineer**
PR.EL.CART. EUROPE ENGINEERING S.R.L. Jun 2025 - Present
Rome, Italy

- Developing advanced software solutions in collaboration with the Italian Ministry of Agriculture
- Managing and mentoring a team of 3 junior developers, guiding them in software architecture
- Independently developed the official **AGEA Vinitaly 2026** application using PMTiles technology
- Engineered an AI pipeline on Google Cloud Vertex AI to generate 1m satellite photos from 10m Sentinel imagery, reducing costs from €4/5M to €500/month

● **Full Stack Software Engineer**
Tiuke S.R.L. Oct 2019 - Jun 2025
Rome, Italy

- Developed 5+ web and mobile applications for the Italian Ministry of Agriculture, serving over 10,000 active users
- Collaborated in a high-performing 5-member development team with shared decision-making, optimizing releases and task coordination
- Refactored legacy monolithic applications into responsive web apps, reducing page load times by 40%
- Created automation tools for integrating mapping layers, accelerating development time by 3x
- Developed backend systems for automated report generation, saving 15 hours of manual work per week
- Implemented specialized LLM agents for agricultural data extraction, achieving 95% accuracy
- Enhanced training of geospatial segmentation models, increasing parcel detection precision by 20%
- Wrote batch scripts to process terabytes of data, reducing pipeline execution time by 50%

Projects

[View more on GitHub](#)

Card Scanner [GitHub: ★ 58] Completed
AI-powered Pokemon card recognition system with client-side processing

- Built a lightweight (<10MB) client-side card identification system that works with all new cards without requiring retraining
- Successfully integrated as the core MVP scanner into the open-source **TCG Pocket Collection Tracker** (★ 150+), significantly enhancing the user experience

EGNSS Capacitor Completed
Universal GNSS Capacitor plugin with OSNMA support

- Developed a cross-platform plugin to guarantee absolute geographic precision in Europe using Galileo's OSNMA and EGNOS
- Exposed a unified API for Android, iOS, and Web to read highly accurate coordinates from internal GNSS and external Bluetooth antennas

Wolf & Mafia Completed
100% Offline Mafia & Werewolf game for Android and iOS

- Developed from scratch a complete offline social deduction game featuring 36 unique roles and a fully guided narrator system
- Powered by a completely deterministic, seed-based game state generator that functions offline without any internet or servers
- Implemented QR-code based local multiplayer allowing users to join games instantly

Game Ecosystem Tools In Progress
ForgeMaster App & Eatventure Seed Predictor

- Developed an auto-updating companion app for ForgeMaster (🌐 / 📱)
- Created an accurate seed-based RNG drop predictor for Eatventure (🌐 / 📱)

Smart House Completed
Arduino-based custom home automation system

- Developed a complete home automation system on Arduino MEGA2560 featuring SSH remote access
- Implemented PWM control for lighting, ADC sensor inputs, and a custom interactive shell interface

UniverCity Completed
Award-winning collaborative platform for university students

- Created an intelligent note-sharing system with advanced indexing algorithms, winning the Google Workshop competition
- Presented the project at Google's offices in Tel Aviv, receiving recognition for innovation and practical application